1. A real estate company wants to develop a system that predicts house prices based on square footage, number of bedrooms, and location.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   Ans: Predict House Prices.
   a. Domain: Machine Learning (House price is Numerical value)
   b. Supervised Learning (As requirement is clear and input and output variables are provided)
   c. Regression Model: As output will be continuous numerical value
   d. Import the required libraries for data processing
   e. Read data file
   f. Use onehotencoder / get_dummies to convert categorical value to numerical value.
   g. Split data into Independent and dependent variable. Here Square footage, number of bedrooms and location are independent variable. House price is output (dependent) variable
   h. Import model_selection library to Split Train dataset and Test dataset
   i. Import regression model and train the model
   j. Predict the model
   k. Evaluate the model using R2_score
   l. If the r_score value is closer to 1 then the model is good.
   m. If not, try other regression model
   n. Once the model is good. Create deployment file using pickle
   o. Test the deployment file

2. A bank wants to build a model to detect fraudulent transactions by analyzing customer spending behavior and transaction history.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   Ans: (Problem statement: Identify Fraudulent transactions)
   a. Domain: Machine Learning (Identify Fraudulent transactions)
   b. Supervised Learning (As requirement is clear and input and output variables are provided)
   c. Classification Model: As output will be Fraud or not.
   d. Import the required libraries for data processing
   e. Read data file
   f. Use onehotencoder / get_dummies to convert categorical value to numerical value.
   g. Split data into Independent and dependent variable.
   h. Import model_selection library to Split Train dataset and Test dataset
   i. Import classification Algorithm and train the model
   j. Predict the model
   k. Evaluate the model using confusion_matrix and classification_report
   l. If the precision value is closer to 1 then the model is good. Here we need to consider precision value than other metrics. To avoid the Type 1 error.
   m. If not, try other classification model

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n. Once the model is good. Create deployment file using pickle
o. Test the deployment file

3. A supermarket wants to segment its customers based on their shopping patterns to provide personalized promotions.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   Ans: (Problem statement: Customer shopping patterns)
   a. Domain: Machine Learning (Identify Shopping patterns)
   b. UnSupervised Learning (As requirement is clear. And there is no output variables provided. Here expectation will be segmentation of customer)
   c. Clustering Model: as customer segmentation
   d. Import the required libraries for data processing
   e. Read data file
   f. Identify the required variables to get pattern and meaningful segmentation
   g. Get those variable as dataframe in an input variable
   h. If the value contains any categorical value, use Labelencoder to convert into Numerical value
   i. Plot the data for checking the data pattern
   j. Import clustering Algorithm and train the model
   k. Get Label value
   l. Append Label value into dataset along with existing dataset
   m. Send the final datafile to stakeholder

4. A company wants to estimate an employee's salary based on their years of experience, job title, and education level.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   Ans: predict employee's salary.
   a. Domain: Machine Learning (Employee salary is Numerical value)
   b. Supervised Learning (As requirement is clear and input and output variables are provided)
   c. Regression Model: As output will be continuous numerical value
   d. Import the required libraries for data processing
   e. Read data file
   f. Use onehotencoder / get_dummies to convert categorical value to numerical value.
   g. Split data into Independent and dependent variable. Here years of experience, job title, and education level are independent variable. Employee's Salary is output (dependent) variable
   h. Import model_selection library to Split Train dataset and Test dataset
   i. Import regression model and train the model
   j. Predict the model
   k. Evaluate the model using R2_score
   l. If the r_score value is closer to 1 then the model is good.

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m. If not, try other regression model
n. Once the model is good. Create deployment file using pickle
o. Test the deployment file

5. An email provider wants to automatically classify incoming emails as spam or not spam based on their content and sender details.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   Ans: (Problem statement: Identify Spam email)
   a. Domain: Machine Learning (Identify Spam email)
   b. Supervised Learning (As requirement is clear and input and output variables are provided)
   c. Classification Model: As output will be Spam or not.
   d. Import the required libraries for data processing
   e. Read data file
   f. Use onehotencoder / get_dummies to convert categorical value to numerical value.
   g. Split data into Independent and dependent variable.
   h. Import model_selection library to Split Train dataset and Test dataset
   i. Import classification Algorithm and train the model
   j. Predict the model
   k. Evaluate the model using confusion_matrix and classification_report
   l. If the Accuracy value is closer to 1 then the model is good.
   m. If not, try other classification model
   n. Once the model is good. Create deployment file using pickle
   o. Test the deployment file

6. A business wants to analyze customer reviews of its products and determine whether the sentiment is positive or negative.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   Ans: (Problem statement: Customer sentiment positive or negative)
   a. Domain: Natural Language Processing (Identify Customer sentiment. All data would be in text.)
   b. I can add the details once I learn Natural Language processing

7. An insurance company wants to predict whether a customer is likely to file a claim in the next year based on their driving history and demographics.
   Q: Identify the problem type and outline the step-by-step logic to solve it.
   Ans: (Problem statement: predict customer future claim.)
   a. Domain: Machine Learning (predict customer future claim.)
   b. Supervised Learning (As requirement is clear and input and output variables are provided)
   c. Classification Model: As output will be customer will claim or not.
   d. Import the required libraries for data processing

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e. Read data file
f. Use onehotencoder / get_dummies to convert categorical value to numerical value.
g. Split data into Independent and dependent variable.
h. Import model_selection library to Split Train dataset and Test dataset
i. Import classification Algorithm and train the model
j. Predict the model
k. Evaluate the model using confusion_matrix and classification_report
l. If the Accuracy value is closer to 1 then the model is good.
m. If not, try other classification model
n. Once the model is good. Create deployment file using pickle
o. Test the deployment file

8. A streaming platform wants to recommend movies to users by grouping them based on their viewing preferences and watch history.
Q: Identify the problem type and outline the step-by-step logic to solve it.
Ans: (Problem statement: to recommend movies to users)
    a. Domain: Machine Learning (to recommend movies to users)
    b. UnSupervised Learning (As requirement is clear. And there is no output variables provided. Here expectation will be segmentation of movies based on user's watch history)
    c. Clustering Model: as movie recommendation
    d. Import the required libraries for data processing
    e. Read data file
    f. Identify the required variables to get pattern and meaningful category
    g. Get those variable as dataframe in an input variable
    h. If the value contains any categorical value, use Labelencoder to convert into Numerical value
    i. Plot the data for checking the data pattern
    j. Import clustering Algorithm and train the model
    k. Get Label value
    l. Append Label value into dataset along with existing dataset
    m. If the any new movie will fall in that identified category then company to recommend this movie to user.

9. A hospital wants to predict the recovery time of patients after surgery based on their age, medical history, and lifestyle habits.
Q: Identify the problem type and outline the step-by-step logic to solve it.
Ans: predict recovery time of patients after surgery.
    a. Domain: Machine Learning (recovery time is numerical value)
    b. Supervised Learning (As requirement is clear and input and output variables are provided)
    c. Regression Model: As output will be continuous numerical value
    d. Import the required libraries for data processing

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e. Read data file
f. Use onehotencoder / get_dummies to convert categorical value to numerical value.
g. Split data into Independent and dependent variable. Here age, medical history, and lifestyle habits are independent variable. Recovery time is output (dependent) variable
h. Import model_selection library to Split Train dataset and Test dataset
i. Import regression model and train the model
j. Predict the model
k. Evaluate the model using R2_score
l. If the r_score value is closer to 1 then the model is good.
m. If not, try other regression model
n. Once the model is good. Create deployment file using pickle
o. Test the deployment file

10. A university wants to predict a student's final exam score based on study hours, attendance, and past academic performance.

Q: Identify the problem type and outline the step-by-step logic to solve it.

Ans: predict final exam score.

a. Domain: Machine Learning (final exam score is numerical value)
b. Supervised Learning (As requirement is clear and input and output variables are provided)
c. Regression Model: As output will be continuous numerical value
d. Import the required libraries for data processing
e. Read data file
f. Use onehotencoder / get_dummies to convert categorical value to numerical value.
g. Split data into Independent and dependent variable. Here study hours, attendance, and past academic performance are independent variable. Final exam score is output (dependent) variable
h. Import model_selection library to Split Train dataset and Test dataset
i. Import regression model and train the model
j. Predict the model
k. Evaluate the model using R2_score
l. If the r_score value is closer to 1 then the model is good.
m. If not, try other regression model
n. Once the model is good. Create deployment file using pickle
o. Test the deployment file

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